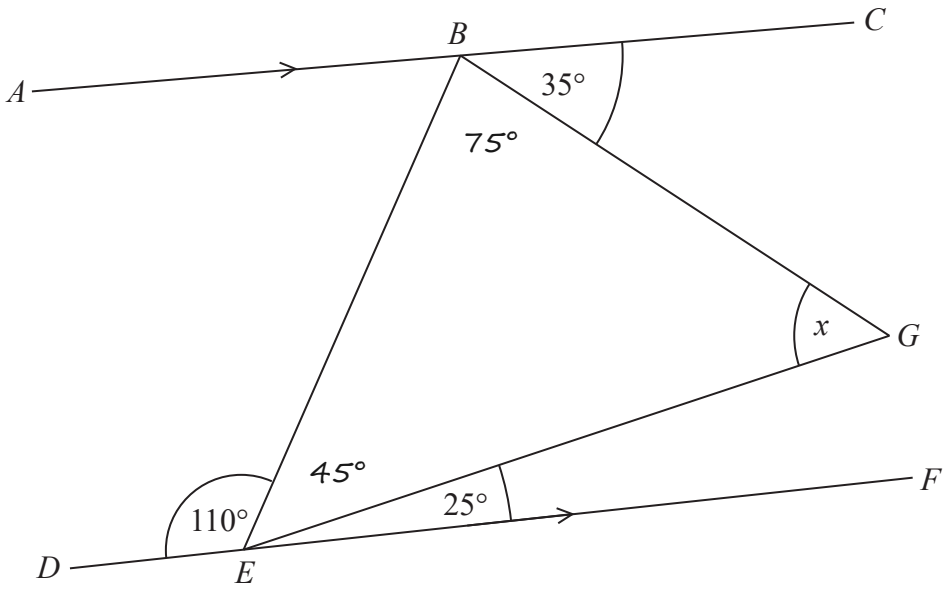
22 *BEG* is a triangle.



ABC and *DEF* are parallel lines.

Work out the size of angle *x*.
Give a reason for each stage of your working.

Angle EBC = 110°

Alternate angles on parallel lines are equal.

1 mark
Use the parallel lines

Angle EBG = 110° – 35° = 75°

1 mark
Linked parallel-line reason and angle-sum reason

Angle BEG = 180° – 110° – 25° = 45°

Angles on a straight line add to 180°.

1 mark
Complete the angle method

x = 180° – 75° – 45° = 60°

Angles in a triangle add to 180°.

1 mark
Correct angle x

60 °

(Total for Question 22 is 4 marks)